

Project Title:
Wind Load Databases in Structural Engineering: A Comparative Literature Review.

Department of Civil and Environmental Engineering (CEE)
Florida International University (FIU)

Project Deadline: 12/01/2024

Submitted by:
Mostafa Sameer, Ph.D.
Student,
Department of Civil Engineering,
Florida International University
(FIU).
msame009@fiu.edu
Contact: 313 335 7658

Submitted to:
Ioannis Zisis, PhD.,
Associate Professor, Co-Director, Lab. Wind
Engineering Engineering Research,
Extreme Events Institute, an FIU Preeminent
Program
Florida International University
Office: EC 3608
Phone: 305.348.4869
Email: izisis@fiu.edu

TABLE OF CONTENTS

ABSTRACT:.....	3
1. Introduction	3
1.1. Overview of Wind Loads and Their Importance in Structural Engineering	3
1.2. Purpose of Wind Load Databases for Engineers	4
1.3. Introduction to the Major Wind Load Databases (TPU and NIST)	4
1.4. Relevance of This Comparative Study	5
2. Literature Review	6
2.1. Previous Studies on Wind Load Databases	6
2.2. Key Findings from Aerodynamic Data in Wind Engineering	6
2.3. Overview of Wind-Related Building Codes and Standards.....	7
3. Wind Load Databases.....	9
3.1. Tokyo Polytechnic University (TPU) Aerodynamic Database	9
3.1.1. Description:	9
3.1.2. Key Features:.....	11
3.1.3. Applications:.....	12
3.2. National Institute of Standards and Technology (NIST) Aerodynamic Database.....	12
3.2.1. Description:	12
3.2.2. Key Features:.....	13
3.2.3. Applications:.....	13
3.3 Contributions to Disaster Resilience	14
4. Comparative Analysis	14
4.1. Data Comparison: High-Rise vs. Low-Rise Buildings.....	15
4.2. Methodologies Used for Determining Wind Pressures	15
4.3. Data Coverage: Geographic Scope and Building Types	16
4.4. Wind-Induced Dynamic Responses	17
5. Conclusion and Recommendations.....	18
5.1. Summary of Key Findings.....	18
5.2. How Wind Load Databases Can Be Improved	19
5.3. Recommendations for Future Research	19
6. Future Research Directions.....	20
7. References	20

LISTS OF FIGURES

Figure 2 Upwind surface roughness conditions for Exposure D ASCE 7 Wind Exposure Categories and How Exposure 'D'	8
Figure 3 Different extreme conditions of structures during natural hazards	9
Figure 4 Different wind load databases in Tokyo Polytechnic University (TPU) Aerodynamic Database	10
Figure 5 Flowchart shows how the TPU Aerodynamic Database contains essential wind tunnel data, categorizes it into different data types, and outlines.	12

LISTS OF TABLES

Table 1 Comparison of Methodologies, Building Types, and Geographic Scope: TPU vs. NIST Databases.....	17
--	----

ABSTRACT:

The study examines various wind load databases, particularly the Tokyo Polytechnic University (TPU) Aerodynamic Database and the National Institute of Standards and Technology (NIST) Aerodynamic Database, which provide essential data for structural engineering. These databases offer comprehensive empirical data, including wind pressure coefficients, aerodynamic forces, and wind tunnel experiment results. Such data are critical for understanding how wind interacts with different building shapes, heights, and configurations. The review also explores the impact of surrounding structures and the variation in wind effects on high-rise buildings. Among the databases, TPU is noted for its detailed data on high-rise buildings, while NIST provides broader coverage for both low-rise and high-rise structures. The paper highlights the importance of these resources for engineers designing resilient buildings, ensuring safety, and meeting building code requirements, such as ASCE 7.

1. Introduction

1.1. Overview of Wind Loads and Their Importance in Structural Engineering

Wind loads refer to the forces exerted by wind on structures, which are crucial for ensuring buildings' safety, stability, and performance. These loads become especially significant for high-rise structures, bridges, and other large infrastructure, as wind-induced forces increase with height and exposure. Accurately assessing wind loads is essential to prevent structural failures, which could otherwise result in severe economic losses, property damage, or even loss of life. Proper wind load evaluation enables engineers to design resilient structures, ensuring they meet safety standards and maintain functionality under extreme weather conditions, such as hurricanes and typhoons.

1.2. Purpose of Wind Load Databases for Engineers

Wind load databases are critical for engineers to design and assess wind-resistant structures. They provide access to a wide range of empirical data, including wind pressure coefficients, aerodynamic forces, and results from wind tunnel experiments, which are essential for understanding how wind interacts with different building shapes, heights, and configurations.

These databases help engineers accurately model the dynamic effects of wind on structures, such as oscillations, vibrations, and pressure variations across building surfaces. For example, in high-rise buildings, the impact of wind is not static but varies with height, building orientation, and environmental conditions like surrounding structures. Databases consolidate complex data on such effects, offering practical insights that engineers can use during preliminary and final design stages.

Additionally, wind load databases help with Database-Assisted Design (DAD), which integrates historical data with real-time simulations to optimize structural performance. Engineers can also use the data to validate compliance with building codes and standards such as ASCE 7, ensuring that structures can withstand both normal and extreme wind events (e.g., hurricanes or typhoons).

These resources not only improve design accuracy but also save time by minimizing the need for individual wind tunnel tests. By leveraging these databases, engineers can predict wind pressures, assess structural safety, and ensure resilience against potential wind hazards, ultimately contributing to safer, more durable infrastructure.

1.3. Introduction to the Major Wind Load Databases (TPU and NIST)

Two prominent wind load databases—Tokyo Polytechnic University (TPU) Aerodynamic Database and the National Institute of Standards and Technology (NIST) Aerodynamic Database—play a key role in modern wind engineering. These databases provide engineers with reliable data to design wind-resistant structures, ensuring safety and durability.

The TPU Aerodynamic Database focuses on high-rise buildings, offering detailed wind tunnel data on wind pressures, interference effects, and dynamic responses. It provides local and area-averaged wind pressure coefficients, which are crucial for understanding how tall structures respond to varying wind forces in dense urban environments. This information enables engineers to predict wind-induced movements and vibrations more effectively, making it particularly valuable for the design of skyscrapers and other large structures.

In contrast, the NIST Aerodynamic Database offers a broader range of data, covering both low-rise and high-rise buildings. NIST's database is especially useful for Database-Assisted Design (DAD), featuring tools like HR_DAD (for high-rise buildings) and wind pressure simulation software, which help engineers predict extreme wind events. The NIST database also includes wind speed analysis tools, which are essential for assessing risks posed by hurricanes, tornadoes, and other extreme weather conditions.

Together, these two databases serve as indispensable resources, providing the empirical foundation needed to design resilient buildings in diverse environments. By offering data for both typical and extreme wind scenarios, they enable engineers to meet regulatory requirements, optimize building performance, and enhance structural safety across a variety of architectural designs.

1.4. Relevance of This Comparative Study

This comparative study of the TPU and NIST aerodynamic databases is essential for identifying each database's strengths, limitations, and practical applications in structural engineering. Both databases offer distinct data sets that are relevant to different types of buildings, making it crucial to understand how engineers can leverage them effectively for diverse design scenarios.

The study is particularly relevant to meeting the requirements of building codes and standards, such as ASCE 7, which govern wind load calculations and ensure the structural safety of buildings. ASCE 7 specifies guidelines for determining wind loads based on building type, height, and location, and this study aims to clarify how the TPU and NIST data align with these standards. For example, TPU's focus on high-rise buildings provides critical insights for urban environments, while NIST's coverage of both low-rise and high-rise structures makes it useful for a wider range of applications, including residential, commercial, and industrial buildings.

By comparing the methodologies, data coverage, and software tools offered by each database, this research provides engineers with a deeper understanding of how to choose the right resource for specific projects. It also highlights potential areas for improvement within these databases, fostering future development in wind load modeling. Ultimately, this study supports the creation of more resilient and sustainable structures, helping engineers apply the most appropriate data to enhance safety, optimize design, and comply with evolving wind load standards.

2. Literature Review

2.1. Previous Studies on Wind Load Databases

Boundary layer wind tunnels have been used in numerous studies to investigate wind loads on structures, and numerical techniques have become more prevalent in recent years. Computational wind load evaluation has become a desirable alternative due to the development of robust sub-grid turbulence models, numerical techniques for creating inflow turbulence that mimics upstream conditions, and advancements in hardware and software technologies (Tamura et al. 2008, Tominaga et al. 2008, Huang and Li 2010, Dagnew and Bitsuamlak 2013). Both low-rise and high-rise structures have been the subject of previous numerical studies. The Texas Tech Building (Senthoooran et al. 2004), the Wall of Wind Test Building (Bitsuamlak et al. 2010), and the Silsoe Cube (Wright and Easom 2003) are a few noteworthy experiments of full-scale, low-rise structures. Nozawa and Tamura (2003) used Large-Eddy Simulation (LES) to forecast time-averaged pressure coefficients for high-rise structures on a 1:1:4 scale, while Tamura et al. (2008) offered recommendations for the Architectural Institute of Japan (AIJ) to use when predicting wind loads numerically.

Furthermore, the aerodynamics of the Commonwealth Advisory Aeronautical Research Council (CAARC) model, including flow patterns, mean, and root-mean-square (rms) pressure coefficients on the building's surface, were examined by Huang et al. (2007) and Braun and Awruch (2009). Because of flow interference, wind flow dynamics become more complex when there are nearby structures (Khanduri et al. 1998). Reynolds-averaged Navier-Stokes (RANS) equations have been used in the majority of numerical research evaluating these interference effects (Zhang and Gu 2008, Lam and To 2006). Replicating the inflow conditions seen in boundary-layer wind tunnel studies is crucial to achieving accuracy in numerical models. With an emphasis on the CAARC model, this work offers a methodical examination of many input turbulence production strategies for LES. The work shows extensive validation through comparisons with wind tunnel data received from RWDI USA LLC (Miramar, Florida) and includes simulations with and without neighboring structures (Dagnew et al. 2009, Dagnew and Bitsuamlak 2010).

2.2. Key Findings from Aerodynamic Data in Wind Engineering

Most of the research on the numerical assessment of wind pressures on structures focuses on basic cubic designs that are exposed to wind that is perpendicular to one of their faces. This is mostly because cubes are an easy form to work with and there are many experimental findings available. In this work, experimental data from four investigations that deal with suburban

exposure (Exposure B in North American wind codes and standards) with power-law velocity profile exponents ranging from 0.21 to 0.27 have been analyzed. For different locations on the building's surface, the testing findings are shown as ranges of values rather than precise numbers. By displaying the findings in this manner, the study makes it possible to compare experimental and numerical scatter, which offers important information for evaluations on design [Castro and Hunt, 1977, Hunt, 1982, Murakami and A., 1988, Stathopoulos and D.-B., 1989].

2.3. Overview of Wind-Related Building Codes and Standards

Wind-related building codes and standards are crucial for ensuring that structures can withstand the forces exerted by wind. In the United States, the American Society of Civil Engineers (ASCE) 7 standard governs the calculation of wind loads on buildings and other structures. ASCE 7 provides guidelines for determining the appropriate wind loads based on factors such as the location of the building, its height, shape, and the surrounding environment. The standard specifies how to account for wind speed, pressure coefficients, and exposure categories, which are vital for designing safe and resilient buildings. ASCE 7 emphasizes the importance of reliable wind pressure data for accurate wind load calculations. This data can be obtained through various methods, including wind tunnel testing and computational models. The TPU and NIST databases, which provide comprehensive wind-related data, align with the guidelines outlined in ASCE 7 by offering empirical data that engineers can use to predict wind-induced forces on structures. These databases provide essential information on wind pressure coefficients, aerodynamic forces, and pressure variations across building surfaces, which are crucial for evaluating wind loads according to ASCE 7 standards.

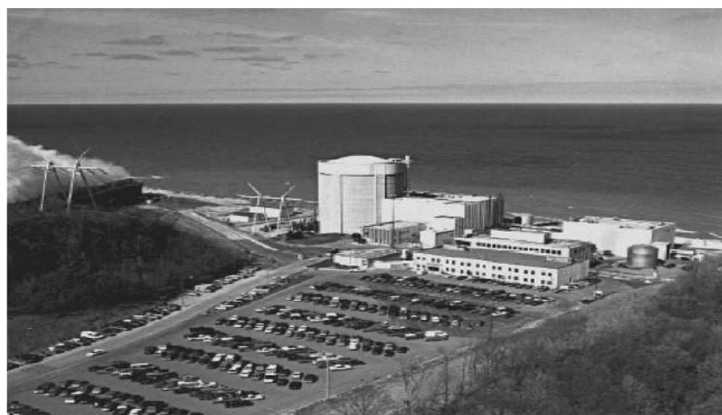


FIGURE C26.7-7 Exposure D: A Building at the Shoreline (Excluding Shorelines in Hurricane-Prone Regions) with wind Flowing over Open Water for a Distance of at Least One Mile. Shorelines in Exposure D Include Inland Waterways, the Great Lakes, and Coastal Areas of California, Oregon, Washington, and Alaska

Figure 1 A building at the shoreline (excluding shoreline in Hurricane-Prone Regions) with wind flowing over open water for a distance of at least one mile, Exposure D) (Ref: Engineering Express. (n.d.). ASCE 7 Wind Exposure Categories and How Exposure 'D' Works.)

The use of exposure categories has been extended by ASCE 7, which clarified a directional need for eight sectors of wind directions. One sector may be exposed to a situation while another is not. The comments went into further detail about this, which was covered in earlier ASCE standards. Sectors and Wind Directions. To calculate the wind loads for each chosen wind direction, the building or structure's exposure must be calculated for the two upwind sectors that extend 45° on either side of the chosen wind direction. Sections 26.7.2 and 26.7.3 will be utilized to determine the exposure in these two sectors, and the exposure that would provide the largest wind loads will be used to depict the winds from that direction.

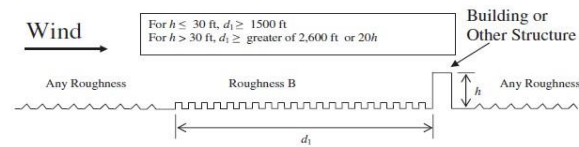


FIGURE C26.7-1 Upwind Surface Roughness Conditions Required for Exposure B

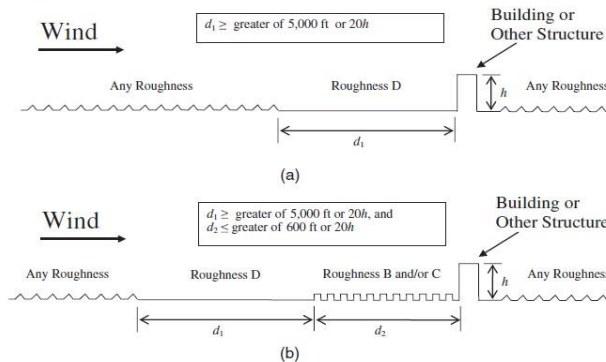


FIGURE C26.7-2 Upwind Surface Roughness Conditions Required for Exposure D, for the Cases with (a) Surface Roughness D Immediately Upwind of the Building, and (b) Surface Roughness B and/or C Immediately Upwind of the Building

Figure 2 Upwind surface roughness conditions for Exposure D (Ref: Engineering Express. (n.d.). ASCE 7 Wind Exposure Categories and How Exposure 'D' Works.)



Figure 3 Different extreme conditions of structures during natural hazards (Ref: Wind Engineering Information Center, n.d.).

3. Wind Load Databases

3.1. Tokyo Polytechnic University (TPU) Aerodynamic Database

3.1.1. Description: When it comes to wind engineering and structural design, engineers and academics rely heavily on the Tokyo Polytechnic University (TPU) Aerodynamic Database. It offers an extensive collection of wind tunnel data, with a concentration on high-rise structures, to help with precise wind load prediction. The database is well known for its comprehensive wind pressure coefficients, which are crucial for comprehending how tall buildings react to different wind forces, especially in crowded cities. Data from wind tunnel tests on several building models, encompassing a broad variety of building heights, forms, and orientations, are included in the TPU Aerodynamic Database. Its focus on capturing the impacts of wind-induced pressure distributions, including mean and variable pressures, on building surfaces is one of its primary characteristics. This information aids engineers in taking into consideration the intricate aerodynamic behavior of tall structures, including dynamic wind-induced motion, vortex shedding, and the overall effect of wind on the performance and stability of a structure.

The TPU database provides information on interference effects that arise from closely spaced buildings in urban settings, in addition to wind pressure data. In real-world settings, where the presence of nearby structures may drastically change the wind flow patterns and pressure distribution, these effects are crucial for precisely estimating wind loads. The TPU database helps

engineers better simulate wind loads by offering both local and area-averaged wind pressure coefficients, which results in safer and more effective building designs.

Building resistance against wind forces is greatly improved by the TPU Aerodynamic Database, which is extensively utilized in wind load modeling for high-rise structures, especially in areas vulnerable to severe winds like hurricanes and typhoons. For engineers designing and evaluating skyscrapers, bridges, and other major infrastructure projects, it is a vital tool for determining whether the structures respect safety regulations and can endure severe weather.

This database facilitates the incorporation of experimental data into design procedures, enabling more precise wind-induced load models and forecasts. The TPU Aerodynamic Database is a vital tool for anybody trying to maximize the performance and safety of structures in wind-prone areas since it provides comprehensive and trustworthy data.

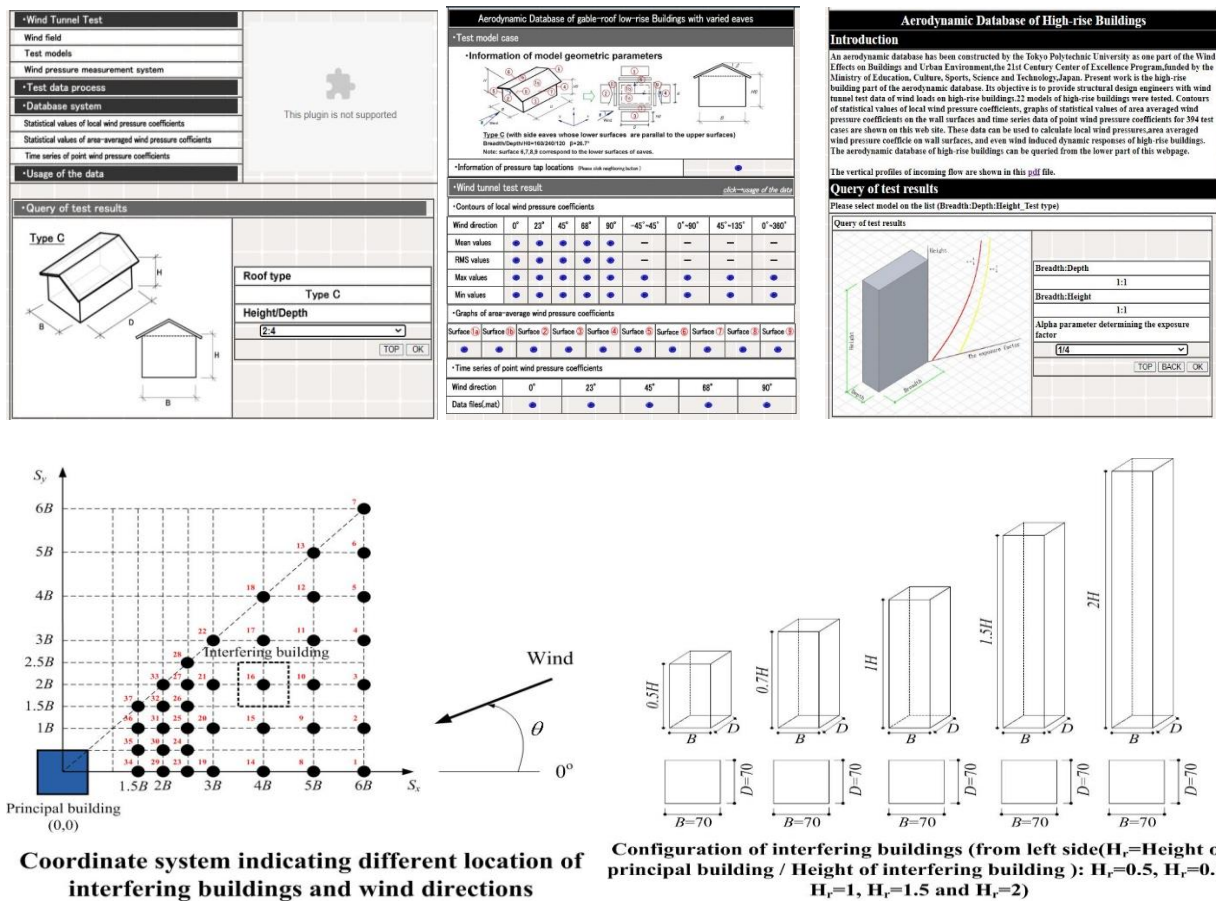


Figure 4 Different wind load databases in Tokyo Polytechnic University (TPU) Aerodynamic Database

3.1.2. Key Features:

- **Wind Tunnel Data on 22 High-Rise Models:** One of the key features of the TPU Aerodynamic Database is its detailed wind tunnel data, which covers 22 different high-rise building models. These models represent various shapes, sizes, and orientations, providing a broad range of wind load scenarios for engineers to analyze. The data includes wind pressure coefficients, both mean and fluctuating, which are essential for accurately predicting wind loads on building surfaces. This diversity in model types allows engineers to simulate a wide array of real-world building designs and assess how they will respond to different wind conditions.
- **Data on Interference Effects and Dynamic Responses:** Another critical aspect of the TPU database is its inclusion of data on wind flow interference effects. When buildings are positioned close together, the wind flow around each building is altered, resulting in complex interactions and changes in the wind pressure distribution. The TPU database captures these interference effects, providing engineers with crucial information on how neighboring structures influence wind loads on a given building. This is particularly valuable in urban environments where buildings are often clustered. Additionally, the database includes data on the dynamic responses of buildings to wind forces, which is important for understanding how a structure will behave under fluctuating wind conditions. This dynamic data helps engineers to predict the potential for wind-induced vibrations and resonances, ensuring that buildings are designed with adequate structural integrity to withstand these forces.

By offering both steady and dynamic wind load data, as well as detailed information on interference effects, the TPU Aerodynamic Database enhances the ability of engineers to create more accurate and reliable wind load models. It supports safer design practices for high-rise buildings, contributing to more resilient infrastructure in wind-prone regions. The database is an essential tool for professionals involved in the design, analysis, and optimization of tall buildings, helping them meet safety standards and improve the overall performance of structures in the face of extreme wind conditions.

The TPU Aerodynamic Database's inclusion of diverse models and its ability to capture complex wind interactions make it a highly valuable resource. Its focus on both the direct wind load and the interference effects between buildings allows for more comprehensive simulations, ultimately leading to better-designed, more wind-resistant structures.

3.1.3. Applications: The flowchart summarizing TPU's wind pressure data structure and its applications. The flowchart shows how the TPU Aerodynamic Database contains essential wind tunnel data, categorizes it into different data types, and outlines its key applications in building design, urban planning, aerodynamic optimization, and research development.



Figure 5 A flowchart shows how the TPU Aerodynamic Database contains essential wind tunnel data, categorizes it into different data types, and outlines.

3.2. National Institute of Standards and Technology (NIST) Aerodynamic Database

3.2.1. Description: The **National Institute of Standards and Technology (NIST)** Aerodynamic Database is a comprehensive resource that provides critical wind pressure data essential for the design and safety analysis of both low-rise and high-rise structures. Established to support the development of building codes and standards, this database plays a key role in ensuring the

resilience of structures against wind loads, which are a significant factor in structural engineering, particularly in regions prone to hurricanes, tornadoes, and high wind events.

For high-rise structures, wind pressure data is especially important. High-rise buildings face significant wind loads at greater heights, where wind speeds are higher and more turbulent. The NIST Aerodynamic Database provides critical data to assess these loads across various design configurations, helping to refine structural designs to withstand lateral forces such as twisting or swaying caused by wind. The database aids in determining the effectiveness of wind-resistant measures, such as aerodynamic building shapes and damping systems, to reduce the impact of wind forces.

3.2.2. Key Features:

The NIST Aerodynamic Database encompasses a wide range of wind pressure data for both **low-rise** and **high-rise** structures. Low-rise structures typically include residential buildings and small commercial buildings, whereas high-rise structures are skyscrapers and other tall buildings. The database includes:

- **Wind Pressure Coefficients (C_p):** These coefficients quantify the pressure exerted by wind on different surfaces of a structure. They vary depending on the shape of the building, the location of the surface (e.g., windward or leeward), and the wind direction.
- **Aerodynamic Properties:** Data related to the aerodynamic behavior of various building shapes under different wind conditions, helping to predict wind-induced forces and moments.
- **Wind Tunnel Test Data:** The database includes results from wind tunnel tests, which simulate real-life wind conditions to measure how buildings react to different wind speeds and directions.

3.2.3. Applications: The data provided by the NIST Aerodynamic Database is crucial for engineers designing buildings that must withstand wind forces. By utilizing the database, engineers can:

- **Ensure Structural Integrity:** Accurate wind pressure data is necessary for calculating load-bearing requirements and ensuring that structures can withstand extreme wind conditions without failing.

- **Enhance Building Codes:** The data supports the development of building codes, such as ASCE 7, which are used worldwide to determine wind loads and other environmental factors affecting building design.
- **Optimize Design Efficiency:** By using precise aerodynamic data, engineers can optimize building designs, reducing material costs and improving overall efficiency without compromising safety.

3.3 Contributions to Disaster Resilience

By improving the understanding of wind-induced forces, the NIST Aerodynamic Database contributes to disaster resilience efforts. Buildings designed with the aid of this data are better equipped to survive severe weather events, minimizing damage, loss of life, and economic impact.

In conclusion, the **NIST Aerodynamic Database** is an essential tool for engineers involved in wind load analysis and building design. Its comprehensive data set, covering both low-rise and high-rise structures, enhances the understanding of wind pressure and aerodynamic behavior, contributing to safer, more resilient buildings. This resource continues to support advancements in wind engineering, building codes, and disaster resilience efforts.

4. Comparative Analysis

TPU Database: With an emphasis on wind resistance and load coefficients for different building types, the TPU (also known as the "Wind-Resistant Design Database") provides crucial wind load information for structures and buildings. It is especially useful for comprehending how various building designs and circumstances impact wind load and how structures interact with wind pressure.

Important attributes: Wind pressure coefficients, wind-induced force calculations, and wind-force-resistant structural design standards are all included in the TPU database. It assists engineers in integrating wind resistance into their building designs by supporting both design codes and real-world applications. It offers information mostly from computational fluid dynamics (CFD) simulations and wind tunnel testing, which is accessible through interactive tools and comprehensive reports.

NIST Aerodynamic Database: The goal of the National Institute of Standards and Technology (NIST) is to provide comprehensive, high-resolution aerodynamic data that is essential for ensuring building safety during windstorms. To enable the wind load estimates required for structures to resist severe weather conditions, NIST created the Aerodynamic Database.

Key Features: Wind pressure coefficients for low-rise and high-rise structures are obtained from field measurements and wind tunnel testing and are available in NIST's database. It provides a wealth of information on wind-induced pressures, wind pressure distribution time series, and how building geometries affect wind load distribution (Yeo et al., 2024).

4.1. Data Comparison: High-Rise vs. Low-Rise Buildings

TPU Database: Focuses primarily on **high-rise buildings**, providing wind load data and pressure coefficients tailored for skyscrapers and similar structures. It offers in-depth analysis for designs that must withstand dynamic wind loads, making it particularly relevant for high-rise engineers looking to model building responses under extreme conditions.

NIST Aerodynamic Database: While **NIST** also includes data on **high-rise buildings**, it stands out by providing a broader scope, including both **low-rise and high-rise structures**. This wider coverage allows it to support various building types in wind load calculations, from residential and commercial low-rise buildings to tall skyscrapers. The NIST database is designed to offer detailed wind pressure coefficients, including data for the full range of wind-induced forces, making it more versatile for engineers working across different types of buildings.

4.2. Methodologies Used for Determining Wind Pressures

TPU Database:

Focus on Dynamic Responses: The TPU Database emphasizes dynamic responses of buildings to wind forces, particularly for high-rise structures. It includes wind tunnel test data and computational fluid dynamics (CFD) simulations that model the dynamic interaction between wind and building surfaces. These tests help assess how a structure's shape and height impact its ability to withstand wind pressure over time, including turbulent wind effects that can lead to vibration or sway. The data is used to inform the design of wind-resistant structures, particularly for skyscrapers and large commercial buildings (Jang et al., 2002, Wind Engineering Information Center, n.d.).

NIST Aerodynamic Database:

- **Focus on Extreme Wind Events:** The NIST database, while also utilizing wind tunnel testing, places more emphasis on **extreme wind events** such as hurricanes and tornadoes. It provides high-resolution **wind pressure coefficients** for both low-rise and high-rise buildings, factoring in data from actual field measurements and controlled laboratory simulations. NIST's methodology is designed to reflect worst-case scenarios,

such as **wind gusts** and **turbulence**, to ensure buildings are designed for resilience in the face of severe weather. The database helps engineers evaluate the aerodynamic effects of extreme wind on buildings' structural integrity (Yeo et al., 2024).

4.3. Data Coverage: Geographic Scope and Building Types

TPU Database:

Geographic Scope: Primarily focuses on urban high-rise environments in Japan, providing detailed wind load data relevant to densely populated areas with tall structures. This makes it particularly valuable for urban planning and high-rise construction, where wind effects are crucial for building stability and design.

Building Types: TPU specializes in high-rise buildings, offering comprehensive data on wind pressures, aerodynamic characteristics, and dynamic responses for tall structures. It supports engineers in designing skyscrapers and other large buildings that must endure significant wind forces.

NIST Aerodynamic Database:

Geographic Scope: NIST provides a broader global perspective, including wind data relevant to both low-rise and high-rise buildings. This data is applicable to a wide range of building scenarios across various geographic regions, including areas susceptible to extreme weather events such as hurricanes and tornadoes.

Building Types: NIST's dataset includes both low-rise and high-rise structures, making it versatile for different types of building designs. It focuses on extreme wind speeds, ensuring that structures are designed to withstand severe weather conditions, and is widely applicable in engineering and construction across the U.S. and internationally.

Table 1 Comparison of Methodologies, Building Types, and Geographic Scope: TPU vs. NIST Databases.

Feature	TPU Database	NIST Aerodynamic Database
Geographic Scope	Urban high-rise environments, primarily in Japan	Global coverage, applicable to various regions, including extreme wind-prone areas
Building Types	Focus on high-rise buildings	Both low-rise and high-rise buildings
Wind Pressure Methodology	Wind tunnel tests and CFD for dynamic responses	Wind tunnel tests, field measurements, and extreme wind events
Focus Areas	Urban wind loads, aerodynamic characteristics of tall buildings	Extreme wind speeds (e.g., hurricanes, tornadoes), wind load data for structural resilience

4.4. Wind-Induced Dynamic Responses

The **TPU Database** provides comprehensive data on **wind-induced dynamic responses** for **high-rise buildings**, with a strong focus on how tall structures behave under fluctuating wind forces. The data covers **aerodynamic effects**, **wind pressures**, and **dynamic behaviors**, such as sway or vibration, that occur in response to wind load variations. This is particularly useful for the design and analysis of skyscrapers and tall buildings, which are subject to significant dynamic wind effects due to their height and exposure (Wind Engineering Information Center, n.d.).

In contrast, the **NIST Aerodynamic Database** primarily focuses on **broader structural responses** and includes data for both **low-rise and high-rise buildings**. While it also includes dynamic wind load data, its emphasis is on extreme wind events (e.g., hurricanes) and the resulting structural resilience. However, NIST's data may not delve as deeply into the specific **dynamic responses** of buildings as TPU's dataset, particularly for very tall buildings (Materials and Structural Systems Division, 2020).

Strengths and Limitations of Each Database

TPU:

Strengths: Offers detailed data on **dynamic wind responses**, including **sway** and **vibration** effects, which are critical for the design of high-rise buildings.

Limitations: Data is mostly focused on **high-rise buildings** and urban environments, which limits its applicability to low-rise or non-urban structures.

NIST:

Strengths: Covers a **wide range of building types**, including both **low-rise and high-rise structures**, and focuses on extreme weather events like **hurricanes** and **tornadoes**, making it valuable for ensuring resilience in severe conditions.

Limitations: May lack the level of detail in **dynamic response** data for **high-rise structures** compared to the TPU database, especially for buildings in highly turbulent wind conditions.

5. Conclusion and Recommendations

5.1. Summary of Key Findings

TPU Database: The Tokyo Polytechnic University (TPU) database is renowned for its comprehensive dynamic response information about urban high-rise structures. Aerodynamic impacts like wind pressures and dynamic behaviors like tall structure sway and vibration are the main topics of this collection. Designing skyscrapers that must endure strong wind forces, particularly in crowded metropolitan settings, requires an understanding of how high-rise buildings react to varying wind forces. Its data, however, does not include a wide range of building types and is mostly restricted to high-rise structures.

NIST Database: By including information on both low-rise and high-rise structures, the NIST Aerodynamic Database provides a more thorough coverage. It is a useful tool for designing structures that must withstand extreme weather conditions since it concentrates on the structural reactions to strong wind events, such as hurricanes and tornadoes. NIST's data offers a more comprehensive view of wind load impacts across different building types and areas than the TPU database, which is more focused on the dynamic wind responses of tall structures. Because of this, it is particularly helpful for comprehending how buildings behave during high-wind events.

5.2. How Wind Load Databases Can Be Improved

- **Expand Datasets to Cover More Climate Variations:** Wind load databases should be extended to include a broader range of **climate conditions**, particularly for areas that are **underrepresented** or have **less severe data**. This could involve including data from regions with **extreme cold** or **tropical environments** to ensure that building designs are suitable for a wider range of **geographical** and **climatic contexts**. Including more **local variations** in wind patterns, such as coastal winds or **mountain winds**, would provide a better foundation for designing structures in diverse environments.
- **Enhance Methodological Frameworks for More Precise Wind Pressure Calculations:** The existing methodologies used for calculating **wind pressures** and **aerodynamic effects** can be refined for more **accurate** and **realistic** predictions. This could involve improving the **precision of computational models** and integrating **advanced computational fluid dynamics (CFD)** methods, which can simulate complex interactions between wind and structures. Additionally, incorporating **real-time data from IoT sensors** could lead to more **responsive models** that consider **dynamic changes** in wind forces during extreme weather events.

5.3. Recommendations for Future Research

- **Improving Data Integration into Engineering Practice:** While wind load databases like TPU and NIST offer valuable data, more work is needed to ensure this data is directly applicable to everyday engineering design. Researchers should explore ways to **simplify the application** of complex wind load data into **building codes** and **design tools**, making it more accessible to engineers and practitioners. This could include developing **user-friendly software** that integrates wind load data into **building design models**, ensuring real-time updates from databases, and facilitating the use of dynamic wind simulations during the design process. Additionally, integrating data from **IoT sensors** and **real-time weather systems** could enhance the accuracy of predictions for specific locations.
- **Updating Standards like ASCE 7:** As wind load data continues to evolve, it is crucial to regularly **update standards** such as **ASCE 7** to reflect the latest research and findings. Future research should examine how new insights from databases like NIST and TPU can inform changes to **building codes** and **wind load provisions** in existing standards. This includes ensuring that updated codes account for **new types of extreme weather events**, **emerging wind patterns**, and the increasing frequency of **climate-related phenomena** like hurricanes and tornadoes. Enhanced data collection methods, such as using

advanced wind tunnel testing or **CFD simulations**, can further improve the accuracy of wind load calculations and help refine these standards.

- **Focusing on Dynamic Wind Loads:** Current standards often overlook the importance of **dynamic wind loads**, particularly for **high-rise buildings**. Research into the **dynamic effects of wind**—how it fluctuates and interacts with different building types—should be prioritized. This could include investigating **wind-induced vibrations**, **building sway**, and **aerodynamic effects** in more detail, particularly in the context of **tall structures** and **urban environments**. Understanding these dynamics will improve the accuracy of wind load assessments and ensure safer designs.

6. Future Research Directions

This study aims to analyze the differences between two major wind load databases: the TPU Aerodynamic Database and the NIST Aerodynamic Database. Both databases play crucial roles in wind-resistant building design, particularly for high-rise structures. The TPU database offers dynamic response data and detailed wind pressure coefficients from wind tunnel tests for tall buildings. It focuses more on dynamic wind loads and responses, making it particularly useful for high-rise structures. In contrast, the NIST database provides comprehensive data across various building types, including both high-rise and low-rise structures, and emphasizes the effects of extreme wind speeds.

Through this analysis, we can gain insights into the methodologies, data collection techniques, and applications of each database. By comparing these databases, this research highlights their strengths and areas where they could be further enhanced. The primary goal of this study is to better understand how these databases can aid engineers in creating safer and more wind-resistant buildings, while also identifying opportunities for improving the wind load assessment process.

7. References

- I.P. Castro, A. G. R. (1977). The flow around a surface-mounted cube in uniform and turbulent streams. *J. Fluid Mech.*, 307–335.
- A. Hunt. (1982). Wind-tunnel measurement of surface pressure on cubic building models at several scales. *J. Wind Eng. Ind. Aerodyn.*, 137–163.
- S. Murakami, A. M. (1988). 3-D numerical simulation of airflow around a cubic model by means of the k-ε model. *J. Wind Eng. Ind. Aerodyn.*, 31, 283–303.
- T. Stathopoulos, M. D.-B. (1989). Design recommendations for wind loading on buildings of intermediate height. *Canad. J. Civil Eng.*, 16(6), 910–916.
- Tamura, T. (2008), “Towards practical use of LES in wind engineering”, *J. Wind Eng. Ind. Aerod.*, 96(10-11), 1451-1471.

- Tominaga, Y., Mochida, A., Murakami, S. and Sawaki, S. (2008), "Comparison of various revised k- ϵ models and LES applied to flow around a high-rise building model with 1:1:2 shape placed within the surface boundary layer", J. Wind Eng. Ind. Aerod., 96(4), 389-411.
- Dagneu, A. and Bitsuamlak, G.T. (2013), "Computational evaluation of wind loads on buildings: a review", Wind Struct., 16(6), 629-660.
- Senthoooran, S., Lee, D. and Parameswaran, S. (2004), "A computational model to calculate the flow-induced pressure fluctuations on buildings", J. Wind Eng. Ind. Aerod., 92(13), 1131-1145.
- Bitsuamlak, G.T., Dagneu, A.K. and Chowdhury, A.G. (2010), "Computational assessment of blockage and wind simulator proximity effects for a new full-scale testing facility", Wind Struct., 13(1), 21-36.
- Wright, N.G. and Easom, G.J. (2003), "Non-linear k- ϵ turbulence model results for flow over a building at full-scale", Appl. Math. Model., 27(12), 1013-1033.
- Nozawa, K. and Tamura, T. (2002), "Large eddy simulation of the flow around a low-rise building immersed in a rough-wall turbulent boundary layer", J. Wind Eng. Ind. Aerod., 90(10), 1151-1162.
- Khanduri, A.C., Stathopoulos, T. and Bedard, C. (1998), "Wind-induced interference effects on buildings-a review of the state-of-art", Eng. Struct., 20, 617-630.
- Zhang, A. and Gu, M. (2008), "Wind tunnel tests and numerical simulations of wind pressures on buildings in staggered arrangement", J. Wind Eng. Ind. Aerod., 96(10-11), 2067-2079.
- Lam, K.M. and To, A.P. (2006), "Reliability of numerical computation of pedestrian-level wind environment around a row of tall buildings", Wind Struct., 9(6), 473-492.
- American Society of Civil Engineers (ASCE). (2021). *Minimum Design Loads and Associated Criteria for Buildings and Other Structures*. American Society of Civil Engineers. <https://doi.org/10.1061/9780784415788>
- Engineering Express. (n.d.). *ASCE 7 WIND EXPOSURE CATEGORIES AND HOW EXPOSURE 'D' WORKS*. <https://www.engineeringexpress.Com/wiki/asce-7-exposure-d-work/>.
- Yeo, D., Pingree, S., Sisson, S., Heumann, E., & Mashnad, M. (2024). *Database-Assisted Design: Performance-Based Approach for Dynamically Sensitive Steel Structures*. <https://doi.org/10.6028/NIST.TN.2293>
- Wind Engineering Information Center. (n.d.). *TPU Aerodynamic Database*. <https://wind.arch.t-kougei.ac.jp/system/eng/contents/code/tpu>.
- Jang, S., Lu, L.-W., Sadek, F., & Simiu, E. (2002). *Database-Assisted Wind Load Capacity Estimates for Low-Rise Steel Frames*. <https://doi.org/10.1061/ASCE0733-94452002128:121594>
- Materials and Structural Systems Division. (2020, January 31). *NIST Aerodynamic Database*. <https://www.Nist.Gov/EI/materials-and-structural-systems-division-73100/nist-aerodynamic-database>.